Chiye Ma

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Research interests: Reinforcement learning, large language models, optimization, manufacturing systems

Education

City University of Hong Kong — Dept. of Mechanical Engineering Sep. 2023 – Present

Ph.D. Candidate in Industrial Engineering & Operations Research

Beijing Institute of Technology — School of Automation Sep. 2020 – Jun. 2023

M.Eng. in Control Engineering

Tianjin University of Science & Technology — College of Automation Sep. 2016 – Jun. 2020

B.Eng. in Measurement and Control Technology & Instrument

Publications

Published Journal Papers

1. **C. Ma**, Z. Jia, J. Zhang, M. Duan, and W. Wang, "Dynamic Modeling and Analysis of Bernoulli Production Lines With Small-Lot Orders and Product Quality Problems," *IEEE Transactions on Industrial Informatics*, 2025.
2. **C. Ma**, T. Yu, H. Chen, X. Pan, J. Zhao, and W. Wang, "A transfer learning-assisted unsupervised clustering approach for unknown fault diagnosis of nuclear power plant equipment," *Progress in Nuclear Energy*, 2027.
3. S. Chen, **C. Ma**, W. Wang, and E. Zio, "An agent-based cooperative co-evolutionary framework for optimizing the production planning of energy supply chains under uncertainty scenarios," *International Journal of Production Economics*, 2024.
4. Z. Jia, Z. Ni, and **C. Ma**, "Dynamic Process Modeling and Real-Time Performance Evaluation of Rework Production System With Small-Lot Order," *IEEE Robotics and Automation Letters (RA-L)*, 2023.
5. J. Zhang, E. Zio, **C. Ma**, K. Liu, and W. Wang, "A probabilistic cost-benefit analysis approach for cyberattack path evaluation," *Reliability Engineering & System Safety*, 2025.

Conference

6. **C. Ma** and W. Wang, "A Lightweight Instruction-Following Language Model for Supply Chain Resilience Management," *4th INFORMS Conference on Quality, Statistics, and Reliability (ICQSR 2026)*, City University of Hong Kong, Hong Kong, Jul. 5–9, 2026.
7. **C. Ma** and W. Wang, "Inventory Management under Disruptions Using Risk-Averse Deep Reinforcement Learning," *2025 INFORMS Annual Meeting*, Atlanta, USA, Oct. 26–29, 2025.
8. **C. Ma** and W. Wang, "Maintenance Decision-Making Optimization for Quality Degrading Production Systems through Deep Reinforcement Learning," *17th International Conference on Probabilistic Safety Assessment and Management & Asian Symposium on Risk Assessment and Management (PSAM17 & ASRAM2024)*, Sendai, Japan, Oct. 7–11, 2024.

Manuscripts

9. **C. Ma**, S. Chen, E. Zio, J. Chen, Z. Jia, and W. Wang, "Language model-assisted rapid production and distribution re-planning under disruptions in energy supply chains," submitted to *International Journal of Production Economics*.
10. **C. Ma**, W. Wang, and M. Xie, "Risk-Averse Multi-Agent Reinforcement Learning for Inventory Management in Single-Manufacturer Multiretailer Systems Under Uncertain Disruptions," manuscript in preparation; target journal: *INFORMS Journal on Computing*.

Research Projects

Risk-Averse Multi-Agent RL for Decision-Making under Uncertain Disruptions Jan. 2022 – Dec. 2024

Core Member • National Natural Science Foundation of China • PI: Prof. Wei Wang, City University of Hong Kong

Extended MAPPO with a sample-based GAE estimator optimizing a CVaR objective rather than expected return, keeping decentralized policies stable in the tail; evaluated on multi-echelon inventory control under stochastic disruptions.

Cooperative Multi-Agent RL for Autonomous Production Control Nov. 2021 – Oct. 2024

Participant • National Key R&D Program Youth Project • PI: Prof. Gang Wang, Beijing Institute of Technology

Formulated adaptive maintenance of degraded machines as a cooperative Dec-POMDP; trained decentralized policies with QMIX value factorization under CTDE.

Transient Performance Evaluation for Complex Production Systems Jan. 2022 – Dec. 2024
Core Member • National Natural Science Foundation of China • PI: Prof. Zhiyang Jia, Beijing Institute of Technology
Derived exact transient dynamics for two-machine one-buffer lines and a recursive algorithm extending them to general serial lines — the Markov-chain models behind the simulation environments used in the RL work above.

Patents

- Z. Jia, **C. Ma**, and Z. Ni, "Transient performance analysis method for production lines with quality problems," Chinese Patent CN114491994B, granted Sept. 17, 2024.
- Z. Jia, Z. Ni, and **C. Ma**, "Transient analysis method for rework systems under customized production," Chinese Patent CN114398782B, granted Sept. 17, 2024.

Honors & Awards

Research Tuition Scholarship, City University of Hong Kong Sep. 2025

Outstanding Academic Performance Award for Research Degree Students, Aug. 2025
City University of Hong Kong

Skills

RL	Multi-agent RL (QMIX, MAPPO), policy-gradient & actor-critic methods, GAE, risk-averse/CVaR objectives, environment design
LLMs	Post-training (domain-specific SFT); data-centric dataset construction, synthesis & filtering
Modeling	Stochastic & Markov-chain modeling, queueing analysis, optimization
Engineering	Python, PyTorch, MATLAB, C; Git, SSH, Linux, Pandas, LaTeX; Docker
Languages	Chinese (native), English (IELTS 7.0, C1)